

Mobile Gym Management System

01 · SYSTEM IDENTITY

What it is: A role-based gym management system - mobile-first (.NET MAUI) + REST API backend + MySQL database - with QR-powered attendance, Gemini AI assistant, and real-time analytics.

What it is NOT: An e-commerce app, social platform, or offline-first app.

Tech Stack

Layer	Technology
Mobile App	.NET MAUI · MVVM Toolkit · CommunityToolkit.Maui
Backend	ASP.NET Core Web API · Entity Framework Core · JWT Auth
Database	MySQL (primary)
Local Storage	SecureStorage (JWT token only)
AI Integration	Gemini API (free tier)
QR Generation	QRCoder library (server-side PNG)

02 · USER ROLES & PERMISSIONS

Two roles: **Admin (Gym Owner)** and **Receptionist (Front Desk Staff)**.

Feature / Action	Admin	Receptionist
Login / Logout	Yes	Yes
Full Dashboard (analytics)	Yes	No
Basic Dashboard (today's stats)	Yes	Yes
View Members	Yes	Yes
Add Members	Yes	Yes
Edit / Delete Members	Yes	No
Manage Membership Plans	Yes	No
View Plans	Yes	Yes
Manage Subscriptions	Yes	No
Record Payments	Yes	Yes
View Full Payment History	Yes	No
View Today's Payments	Yes	Yes
Manual Check-in / Check-out	Yes	Yes
QR Scan Check-in	Yes	Yes
AI Assistant	Yes	No

Feature / Action	Admin	Receptionist
Reports Module	Yes	No
Audit Log	Yes	No
System Settings	Yes	No

03 · SUBSCRIPTION LIFECYCLE (State Machine)

State	Trigger / Condition
PENDING	Subscription created, StartDate not yet reached
ACTIVE	StartDate reached
EXPIRING SOON	EndDate - Today <= 7 days - triggers local notification
EXPIRED	EndDate passed
RENEWED (new row)	New payment recorded -> new Subscription row created

Note: Status is computed on-read by the API - no background cron job required for MVP.

Note: Renewal always creates a NEW Subscription row. Old record stays EXPIRED for history.

Note: New subscription StartDate = day after previous EndDate (if still active), else today.

04 · DATABASE SCHEMA (9 Tables)

Users

UserID (PK) · Username · PasswordHash · Role (Admin/Receptionist) · FullName · IsActive · CreatedAt

Members

MemberID (PK, UUID) · FirstName · LastName · Gender · Birthdate · Phone (unique) · Email (unique, nullable) · Address · EmergencyContactName · EmergencyContactPhone · ProfilePhotoUrl · QRToken (unique UUID - not MemberID) · Status (Active/Inactive) · DateRegistered · CreatedByUserID (FK -> Users)

MembershipPlans

PlanID (PK) · PlanName · Price (decimal) · DurationDays (int) · Description · IsActive (bool - soft delete) · CreatedAt

Subscriptions

SubscriptionID (PK) · MemberID (FK) · PlanID (FK) · StartDate · EndDate (auto = StartDate + DurationDays) · Status (Pending/Active/ExpiringSoon/Expired) - computed on-read · PaymentID (FK) · CreatedByUserID (FK) · CreatedAt

Payments

PaymentID (PK) · MemberID (FK) · PlanID (FK) · Amount (decimal) · PaymentMethod (Cash/GCash/Maya/Bank) · ReferenceNumber (nullable) · DatePaid · Notes (nullable) · RecordedByUserID (FK) · CreatedAt

Attendance

AttendanceID (PK) · MemberID (FK) · CheckInTime (datetime) · CheckOutTime (datetime, nullable) · Mode (Manual/QR) · RecordedByUserID (FK) · Date (date - for fast daily queries)

ProgressRecords (optional)

RecordID (PK) · MemberID (FK) · Weight · Height · BMI · BodyFatPercent · Notes · RecordedAt

AuditLogs

LogID (PK) · UserID (FK) · Action (e.g. MEMBER_CREATED) · EntityType (Member/Payment/etc.) · EntityID · Details (JSON snapshot) · Timestamp

ChatLogs (optional)

LogID (PK) · UserID (FK) · Message · Response · Intent · Timestamp

05 · REST API ENDPOINTS

Auth

Method	Endpoint	Notes
POST	/api/auth/login	Returns JWT + role
POST	/api/auth/logout	Invalidate token
POST	/api/auth/refresh	Refresh JWT

Members

Method	Endpoint	Notes
GET	/api/members	Paginated, filterable list
GET	/api/members/{id}	Member detail
POST	/api/members	Create member + generate QRToken
PUT	/api/members/{id}	Update member (Admin)
DELETE	/api/members/{id}	Soft delete (Admin)
GET	/api/members/{id}/qr	Returns QR PNG image
GET	/api/members/{id}/subscriptions	Subscription history
GET	/api/members/{id}/payments	Payment history
GET	/api/members/{id}/attendance	Attendance history

Plans

Method	Endpoint	Notes
GET	/api/plans	List active plans
GET	/api/plans/{id}	Plan detail
POST	/api/plans	Create plan (Admin)
PUT	/api/plans/{id}	Update plan (Admin)
DELETE	/api/plans/{id}	Soft delete (Admin)

Subscriptions

Method	Endpoint	Notes
GET	/api/subscriptions	List, filterable by status
GET	/api/subscriptions/{id}	Detail
POST	/api/subscriptions	Create - auto-creates linked payment (tx)
PUT	/api/subscriptions/{id}	Update (Admin)

Payments

Method	Endpoint	Notes
GET	/api/payments	Filterable by date, method, member
GET	/api/payments/{id}	Detail
POST	/api/payments	Record payment

Attendance

Method	Endpoint	Notes
GET	/api/attendance	Filterable by date / member
POST	/api/attendance/checkin	Open attendance record
POST	/api/attendance/checkout	Close attendance record
GET	/api/attendance/today	Today's list

QR System

Method	Endpoint	Notes
POST	/api/qr/scan	Body: { token } -> returns MemberProfile + ActiveSubscription + Attendance

AI Chat

Method	Endpoint	Notes
POST	/api/chat/query	Body: { message } -> returns Gemini-formatted response

Reports (Admin)

Method	Endpoint	Notes
GET	/api/reports/revenue	Date range + method filter
GET	/api/reports/attendance	Date range + member filter
GET	/api/reports/members	Status filter
GET	/api/reports/expiring	Days-until-expiry filter

Dashboard

Method	Endpoint	Notes
GET	/api/dashboard/admin	Full analytics snapshot
GET	/api/dashboard/receptionist	Today's stats only

Audit (Admin)

Method	Endpoint	Notes
GET	/api/audit	Paginated log of all user actions

06 · QR ATTENDANCE - DETAILED FLOW

QR Token Rules

- QRToken is a UUID v4 - separate from MemberID (security best practice)
- Generated automatically on member creation by the API

- Stored in Members table as a unique indexed column
- Exposed via GET /api/members/{id}/qr as a server-rendered PNG image
- Displayable in-app on Member Profile screen + printable via MAUI share sheet

Scan -> Check-in Flow

Step	Actor	Action
1	Receptionist	Opens QR Scanner screen in MAUI app
2	MAUI Camera	Decodes QR -> extracts QRToken string
3	MAUI -> API	POST /api/qr/scan { token: 'uuid' }
4	API	Looks up Member by QRToken
5	API -> MAUI	Returns: MemberProfile + ActiveSubscription + AttendanceStatus
6	MAUI UI	Displays Member Card (photo, name, plan, expiry status)
7	Receptionist	Reviews card - taps Confirm Check-in button
8	MAUI -> API	POST /api/attendance/checkin { memberId, mode: 'QR' }
9	API	Creates Attendance record, returns success
10	MAUI UI	Shows success toast - ready for next scan

Note: Profile preview before confirmation prevents accidental check-ins - cleaner UX and stronger defense demo.

07 · AI ASSISTANT - LOCKED QUERY VOCABULARY

Architecture: User message -> API intent matching -> pre-built DB query -> context string -> Gemini API prompt -> natural language response -> MAUI chat UI.

Anything outside the 15 locked intents -> Gemini responds: "I can only answer gym-related questions."

#	User Says...	Maps To
1	How many active members	COUNT active subscriptions
2	Who expires this week	Members with EndDate <= 7 days
3	Revenue this month	SUM payments for current month
4	Revenue today	SUM payments for today
5	How many attended today	COUNT attendance for today
6	Who checked in today	LIST attendance for today
7	Member info [name]	GET member by name search
8	When does [name] expire	GET subscription EndDate
9	Is [name] active	GET member subscription status
10	How many members total	COUNT all members
11	New members this month	COUNT members registered this month
12	Most active members	TOP 5 by attendance count
13	Unpaid or expired members	LIST expired subscriptions
14	Payment history [name]	GET payments by member
15	Plans available	LIST all active plans

08 · REPORTS MODULE

Report	Filters	Output
Revenue Report	Date range, payment method	Total + breakdown by method + daily chart
Attendance Report	Date range, member	Count per day, peak hours
Member Status Report	Status filter	List with plan + expiry date
Most Active Members	Date range	Top 10 by visit count
Expiring Memberships	Days until expiry	Sorted list with contact info

Note: All reports: table view + summary cards. Optional CSV export via MAUI share sheet.

09 · LOCAL NOTIFICATIONS

Implemented via CommunityToolkit.Maui. No external email/SMS service required.

Trigger Condition	Notification Message	When Fired
EndDate - Today <= 7 days	[Warning] {N} memberships expiring soon	On every app launch
EndDate - Today <= 1 day	[Alert] {MemberName} expires tomorrow	On every app launch

10 · SCREEN MAP

Auth

- Login Screen

Admin Navigation

- Admin Dashboard (full analytics)
- Members List -> Member Detail -> Add/Edit Member
- Member Detail sub-screens: Subscription History, Payment History, Attendance History
- Member Detail: QR Code display + print/share
- Plans Management -> Add/Edit Plan
- Subscriptions List
- Reports Screen (5 report types)
- AI Assistant Chat Screen
- Audit Log Screen
- Settings Screen

Receptionist Navigation

- Receptionist Dashboard (today only)
 - Members List (view + add only)
 - Manual Check-in / Check-out Screen
 - QR Scanner -> Member Card Preview -> Confirm Check-in
 - Record Payment Screen
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11 · DEFENSE READINESS CHECKLIST

Area	Status	Evidence
Real database design (9 tables)	Done	Section 04
REST API (35+ endpoints)	Done	Section 05
Mobile architecture (MVVM)	Done	.NET MAUI + MVVM Toolkit
JWT Authentication	Done	Section 01 / Auth endpoints
Role-based access control	Done	Permission matrix - Section 02
CRUD completeness	Done	All modules covered
Subscription lifecycle logic	Done	State machine - Section 03
QR automation (scan + profile)	Done	Section 06
AI integration (Gemini)	Done	15 locked intents - Section 07
Reports module	Done	Section 08
Local notifications	Done	Section 09
Audit/activity log	Done	AuditLogs table - Section 04
Offline mode	Removed	Intentional - avoids sync complexity